

SARAH AL-MAHMOUD

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SUMMARY

Backend-focused AI Engineer with hands-on experience building retrieval-augmented generation (RAG) pipelines, LLM-integrated APIs, and production-grade data services. Recent Computer Science graduate with a strong foundation in distributed systems, vector search, and applied machine learning, delivered through two internships and three end-to-end AI projects. Proven ability to ship reliable, well-tested backend systems in Python and Node.js under tight deadlines. Seeking an Entry-Level AI/Backend Engineer role to build scalable, intelligent applications.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, Java, C++

AI/ML & LLM Tooling: LangChain, LlamaIndex, OpenAI API, Anthropic API, Hugging Face Transformers, scikit-learn, PyTorch, Pinecone, FAISS, ChromaDB

Backend & Infrastructure: FastAPI, Flask, Node.js, Express, REST APIs, PostgreSQL, MongoDB, Redis, Docker, AWS (EC2, S3, Lambda), CI/CD (GitHub Actions)

Core Concepts: Retrieval-Augmented Generation, prompt engineering, embeddings & vector search, microservices, unit/integration testing, Agile/Scrum

EXPERIENCE

Backend Engineering Intern — FlyRank

Jun 2025 – Present

Dubai, UAE

- Built and shipped a Python/FastAPI microservice that classifies and matches website images to content topics using CLIP embeddings, reducing manual content-tagging effort by 70% across client sites.
- Designed a retrieval layer over a 40,000+ document knowledge base using ChromaDB and OpenAI embeddings, cutting average query latency from 1.8s to 420ms through batching and caching.
- Implemented usage metering and cost-tracking middleware for internal LLM API calls, enabling accurate per-client billing across three product lines.
- Wrote unit and integration tests (pytest) achieving 88% coverage on new services; set up automated test runs in GitHub Actions.

Software Engineering Intern — Careem

Jun 2024 – Aug 2024

Dubai, UAE

- Developed a Node.js/Express internal tool that automated weekly driver-incentive reports, saving the operations team approximately 6 hours per week.
- Optimized 4 slow PostgreSQL queries feeding an internal analytics dashboard, reducing average page load time by 35%.
- Partnered with the data team to build a validation pipeline that caught malformed records before ingestion, reducing downstream data errors by 22%.

PROJECTS

JobRadar UAE — Multi-Agent Job Search System | Python, LangChain, FastAPI, PostgreSQL, React

- Designed a multi-agent pipeline (scraper, matcher, and ranker agents) that aggregates listings from 5 UAE job boards and ranks them against a user's resume using embedding similarity.
- Built a RAG-based resume Q&A feature allowing users to ask natural-language questions about their own resume, powered by a FAISS vector store and the Anthropic API.
- Deployed the system on AWS EC2 with a Dockerized FastAPI backend and a React frontend, currently processing 1,000+ job listings weekly.

LLM Usage Metering & Billing Engine | Python, FastAPI, Redis, PostgreSQL, Stripe API

- Built a token-level usage tracking engine for multi-tenant LLM applications, logging cost per request across OpenAI and Anthropic models in real time.
- Implemented tiered billing logic and automated monthly invoicing via the Stripe API, supporting 3 pricing tiers.
- Load-tested the system to reliably handle 500 requests/second using Redis-backed rate limiting and async request queues.

AI Image Understanding & Content Matching Engine | Python, PyTorch, CLIP, FastAPI

- Fine-tuned a CLIP-based model to classify website images into 25 content categories with 91% validation accuracy.
- Built an image-to-text matching API that recommends relevant on-page content based on visual similarity, integrated into FlyRank's internal SEO toolset.

EDUCATION

American University of Sharjah (AUS) — B.S. in Computer Science

Jun 2026

Sharjah, UAE | GPA: 3.81/4.0 | Dean's List: 6 semesters

- Relevant Coursework: Machine Learning, Distributed Systems, Database Systems, Data Structures & Algorithms, Cloud Computing, Natural Language Processing

CERTIFICATIONS

AWS Certified Cloud Practitioner (2025) • DeepLearning.AI — LangChain for LLM Application Development (2025) • Meta Back-End Developer Professional Certificate (2024)